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Received: 9 September 2023 | Revised: 21 October 2023 | Accepted: 5 November 2023

DOI: 10.1002/spy2.354

RESEARCH ARTICLE

WILEY

Safeguarding the IoT: Taxonomy, security solutions, and future research opportunities

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Abstract

With the advancement of innovative technologies, the Internet of Things (IoT) has become quite popular in various applications such as smart homes, smart cities, agriculture, healthcare, and so forth. However, the heterogeneity of IoT protocols can pose security challenges to wireless networks. Thus, we have presented a systematic review of IoT technologies such as ZigBee, Radio Frequency Identification (RFID), Bluetooth, WiFi, Ethernet, AR, embedded subscriber identification modules (E-SIM), and smart sensors, along with their communication protocols, which are part of the IoT-layered architecture. We have bifurcated the IoT-layered architecture into the perception, network, and application layers. Furthermore, we have presented IoT security attacks associated with the perception, network, and application layers of the IoT-layered architecture. Moreover, we have discussed several IoT security solutions, open issues, and research challenges dedicated to the security of IoT networks. IoT security solutions can provide secure and efficient data communication over wireless networks.

KEYWORDS

communication protocols, internet of things, security attacks, security framework

1 | INTRODUCTION

The Internet of Things (IoT) is widely embedded in sensor devices, technology, software, and so forth. The IoT connects various objects with the Internet, like cars, machines, day-to-day life appliances, and so forth. The IoT acts as the mediator. It receives the information from the sender device and sends it to the server or host, where many algorithms process it. This process information is sent to the intermediate, and the end user receives the command. This is very useful nowadays because of its low cost, low power consumption, good connectivity, and many more. Because of its vast working capabilities, it is also used in industries to manufacture different goods, monitor processes, switch products, and so forth. The main benefit in the industry is that the cost of production reduces and further increases production efficiency, and so forth. On a deeper note, four parts come under the domain of IoT framework: sensors, data processing, connectivity, and user interface. Nowadays, IoT applications are used majorly for controlling IoT technologies. IoT applications are software as a Service-based application that helps respond to the IoT device. These applications collect the data from the IoT devices and help in analyzing the data. Many researchers have presented an extensive review on IoT communication protocols and architecture.[1-3]

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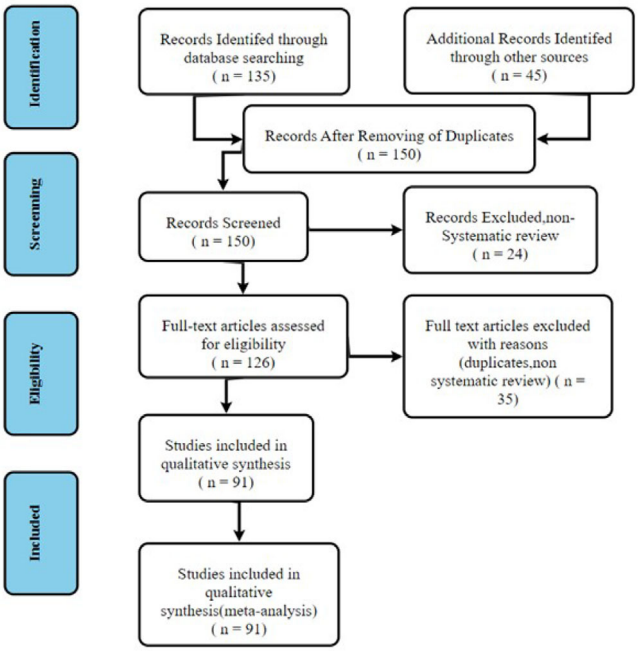

FIGURE 1 Prisma diagram of the proposed survey.

In this regard, Parul Datta et al.[4] presented a review of IoT protocols and five-layer IoT architecture. They also discussed a smart city-based application of IoT by showing the comparison between communication protocols. They briefly discussed some of the protocols without adequately defining them. Then, Poonam Gupta et al.[5] studied widely used application layer protocols in their study. They mentioned their characteristics, suitability, and implementation setup for the following protocols. They concluded their study by comparing these protocols based on practical factors like real-time application, security, power consumption, and so forth. They only talked about Application layer protocols with their implementation. Later, the authors of Reference 6 talked briefly about IoT technologies, IoT five-layered architecture, and protocols. However, they did not mention the challenges to technologies and protocols caused by security attacks. Further, Ahmad Roihan et al.[7] had briefed about IoT three-layer architecture and their respective protocols. However, they have not considered security measurements of IoT architecture similar to the survey proposed by Reference 6. Then, the authors in Reference 8 have extensively surveyed the application layer protocols. They have included the text communication methods protocols that may be used while communicating.

Considering the outlook of literature, most researchers have surveyed the IoT protocols and architecture without considering the IoT technologies and the associated security challenges that can cause them to be vulnerable to the confidential data in the network. Thus, motivated by the aforementioned research gaps, we have proposed a comprehensive survey on IoT technologies and their communication protocols in an IoT-layered architecture. We evaluated security threats on each layer as well as common security attacks on the entire system within the IoT-architecture. Moreover, we have discussed various IoT security solutions to secure the network against any malicious attack. In this paper, we have identified 130 records through database searching and an additional 40 records through other sources on the Internet. We have removed 20 duplicate articles from these, which further results in 150 remaining papers. Further, 30 papers were excluded and found to have a non-systematic review. From the remaining 120 papers, we assessed them fully and removed around 35 papers due to duplicacy and non-systematic review issues. So, in the end, we have considered 85 papers for the proposed survey. Figure 1 shows the Prisma diagram of the proposed survey to show the relevant research papers considered for the comprehensive survey.

1.1 | Scope of the survey

Many researchers have studied the security issues in IoT networks but have yet to consider the communication protocols, security issues, and technologies simultaneously involved in the IoT networks. For example, Kholoud Y. Najmi

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et al.[9] mentioned security issues of the IoT systems with significant applications of IoT, like smart homes and smart cities. They also discussed protocols and architecture of IoT systems with the appropriate security tools to ensure a safe environment for IoT users. It also presented countermeasures based on particular attacks on the layered architecture of IoT devices. Mariam Alhawamdeh et al.[10] added modern discoveries and solutions for the security of IoT applications using the latest technologies. They provided a survey study of current techniques to integrate IoT with modern technologies like Blockchain, Artificial Intelligence, Machine Learning, SDN, and Lightweight systems to aid the security services for the system. Yugha et al.[11] analyzed numerous research challenges and security issues of IoT protocols. They also presented various simulators for the analysis of protocols under IoT. They summarized IoT protocols and security attacks based on IoT devices' three-layer architecture structure. They also focused on research trends of IoT protocols. Soraya Sinche et al.[12] presented an overview of protocols, frameworks, and technologies for managing IoT devices. Their analysis contains comparative views, a standardization timeline, and an analysis of the market. These management protocols, technologies, and frameworks for IoT devices should consider energy efficiency, complexity, processing, and latency. Jie Lin et al.[13] reviewed IoT architectures, technologies, security issues, and combining fog/edge computing and IoT to support different applications. They also discussed possible architectures of IoT devices, which include the conventional three-layer architecture and the service-oriented four-layer architecture. Surapon Kraiajk et al.[14] describes the evolution and importance of IoT in daily life, the architecture of IoT devices, its most widely used protocols, various applications, security and privacy concerns in IoT, real-world implementation of IoT system by using Arduino and its future trends. Vikas Hassija et al.[15] reviewed the security challenges and threats categorized by layer architecture of IoT systems. After discussing the security issues, various emerging technologies focused on achieving high efficiency and output in IoT applications are discussed. They also discussed four technologies: Blockchain, Fog computing, Edge computing, and Machine Learning to increase the security in IoT systems. Afrah Salman Dawood[16] studied various stages and applications of IoT platforms. He also gave an overview of the challenges of IoT systems. Nataliia Neshenko et al.[17] provided an exhaustive classification of surveys concerning various aspects of the IoT system. It helps future IoT research by comparing and contrasting numerous research contributions. Subsequently, they provided a unique taxonomy, which helps to understand IoT weaknesses, their attack vectors, effects on security objectives, attacks that exploit such vulnerabilities, countermeasures methodologies, and currently offered operational security capabilities to summarize and monitor such weaknesses. Table 1 shows the comparative analysis of various IoT security surveys with the proposed survey.

1.2 | Analysis of paper

Figure 2 represents the research papers considered for the proposed survey for the authors. So, it is observed from the figure that we have considered a maximum number of papers from Shah Munam Ali and Conti Mauro. Figure 3 shows the amount of studies we have considered for our review from various sources. Our study sources mainly consist of proceedings and articles with zero collection from miscellaneous and collection sources. Figure 4 decides the keywords most used in the titles of the studies we have reviewed from many authors. These can help to understand the portion of the topic we have discussed in our paper of specific keywords. Figure 5 represents the year-wise number of research papers we have included. The conclusion made from the plot is that a large proportion of papers are taken from recent years, with the maximum number from 2017 to 2020. This is because, in recent years, many developments have taken place in this field and the increasing rate of users due to its increasing requirement. Some papers taken from past years state the basic knowledge of IoT and its usage.

1.3 | Research contributions

The significant research contributions are listed as follows:

1. We have proposed an in-depth survey on IoT technologies and their communication protocols based on the considered IoT-layered architecture.
2. The IoT-layered architecture is mainly classified into three layers, the perception layer, network layer, and application layer, to represent further the security attacks associated with each layer.
3. Moreover, we have discussed IoT security solutions and corresponding open issues and research opportunities of IoT networks.

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TABLE 1 Comparative analysis of various state-of-the-art IoT security surveys with the proposed survey.

Author	Year	Objective	Pros	Cons
Najmi et al.[9]	2021	Presented a comprehensive survey on security and privacy issues in IoT along with the potential risks and possible tools required to secure the data communication between users with confidentiality and reliability.	Security is improved in terms of reliability, integirty and confidentiality	various security threats such as data manipulation, sniffing, data spoofing, etc needs to be considered
Alhawamdeh et al.[10]	2021	Studied an exhaustive survey on new approaches and solutions proposed by the researchers in the field of security in Internet of Thing	Ensure security and privacy of the data in IoT to some extent	AI leveraging and SDN based control and monitoring are ignored
Yugha et al.[11]	2020	Discussed a comprehensive survey from origin of network and various layer of the IoT network till security attack and the countermeasure in IoT	Efficient, lightweight security and trust management protocol are included	Efficiency, Lightweight security and trusted protocol is not focused
Sinche et al.[12]	2020	Given an comprehensive survey on taxonomy of IoT technologies, protocols and IoT device management	Complexity, energy, latency of IoT systems are taken in consideration	Security and privacy of the data are ignored
Lin et al.[13]	2017	Presented an exhaustive survey on IoT architecture, technology, security, privacy issue and edge computing	Case study of real time scenario is discussed	Latency and complexity of IoT devices are ignored
Kraiajk et al.[14]	2015	Given an comprehensive survey on evolution, importance, architecture, protocols and application of IoT	Lightweight and fast communication protocol is designed in this paper	Security and permission policy for IoT are not focused
Hassija et al.[15]	2019	Discussed a detailed survey on security challenges, solutions and technologies in IoT	Data complaince, safety and bandwidth issues are focused	Scalability, availability of blockchain, security of netwrok are ignored
Dawood et al.[16]	2020	Presented a comprehensive survey on various applications, architecture, platforms, challenges in IoT	Inclusion of various application of IoT	No effort on discussion of malware and ransomware detection and data assurance is not focused
Neshenko et al.[17]	2019	Given an exhaustive survey on vulnerability and security in IoT	Minimize energy consumption, energy production, personal security are focused	Complexity and mobility weaknesses are not considered
The proposed survey	2023	Proposed an extensive survey on IoT technologies and communication protocols	Considered aspects of IoT networks to ensure security	-

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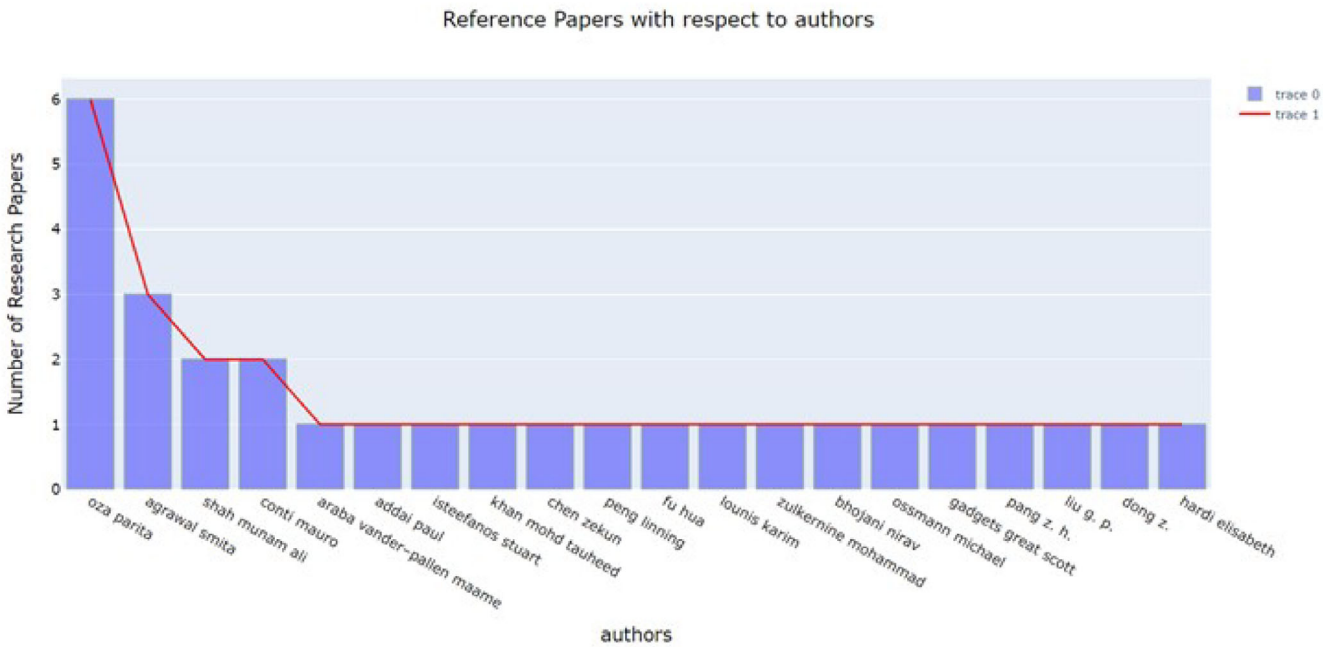

FIGURE 2 Research papers based on authors.

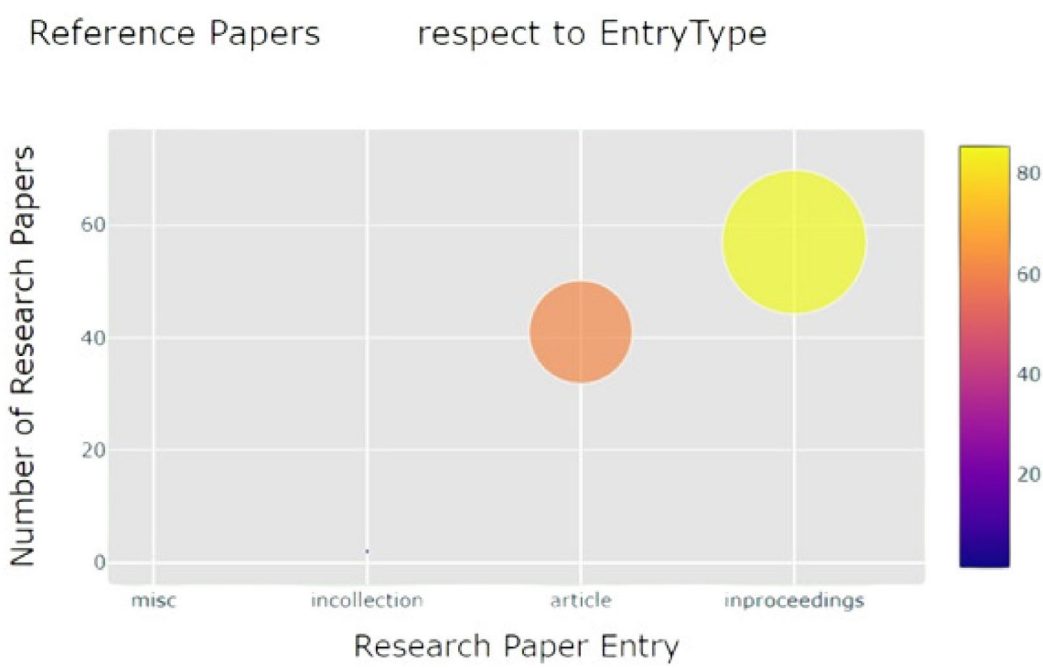

FIGURE 3 Reference papers concerning entry type.

1.4 | Organization

The rest of the paper is represented as follows. Section 2 discussed the various IoT technologies and IoT-layered architecture. Section 3 shows the taxonomy of IoT security attacks on IoT networks. Section 4 represents the IoT security solutions. Section 5 highlights the open issues, research challenges, and opportunities for IoT networks. Finally, the paper is concluded in section 6. Furthermore, Figure 6 represents the sequence flow of the proposed survey, which initiates with the IoT background and ends with the conclusion.

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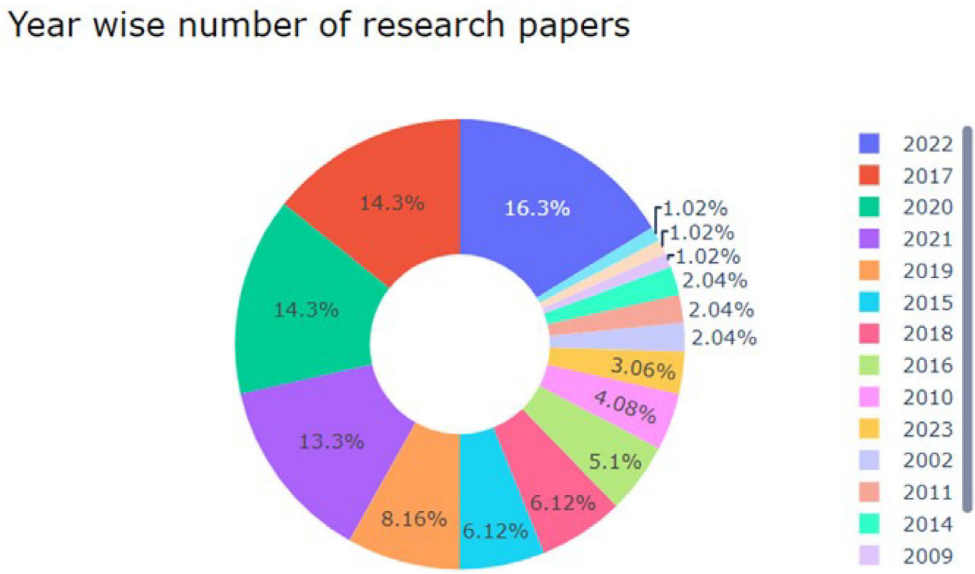

FIGURE 4 Reference papers concerning keywords.

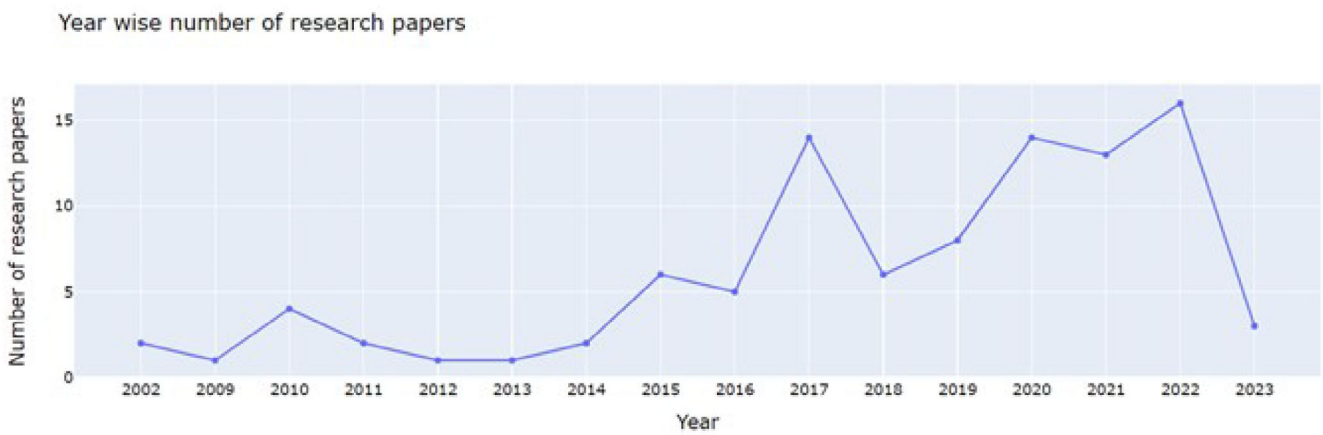

FIGURE 5 Year-wise number of research papers.

2 | BACKGROUND CONCEPTS

2.1 | IoT technologies

2.1.1 | Zigbee

Zigbee is a low-cost, low-power technology that helps in communication within small areas (around 20 to 25 m) with the help of wireless ad hoc technology. This technology works on mesh topology. It works on IEEE 802.15.4 standard. This standard was developed by Zigbee Alliance, who envisioned this in the early 1998s. Many specifications have been validated in this technology; in December 2004, the IEEE 802.15.4 standard was ratified; in June 2005, specification 1.0 was announced, which was named as Zigbee 2004 Specification and further was renewed as Zigbee 3.0 in 2016.[18] Zigbee standard is typically built to control sensor networks. There are three ranges of frequency assigned to the Zigbee currently. For example, Europe has 868MHz, the USA and Australia have 915 MHz, and 2.4 GHZ worldwide. In the network layer, Zigbee routes different types of information through Ad hoc On-Demand Distance Vector Routing. There are three components in the Zigbee device as follows:

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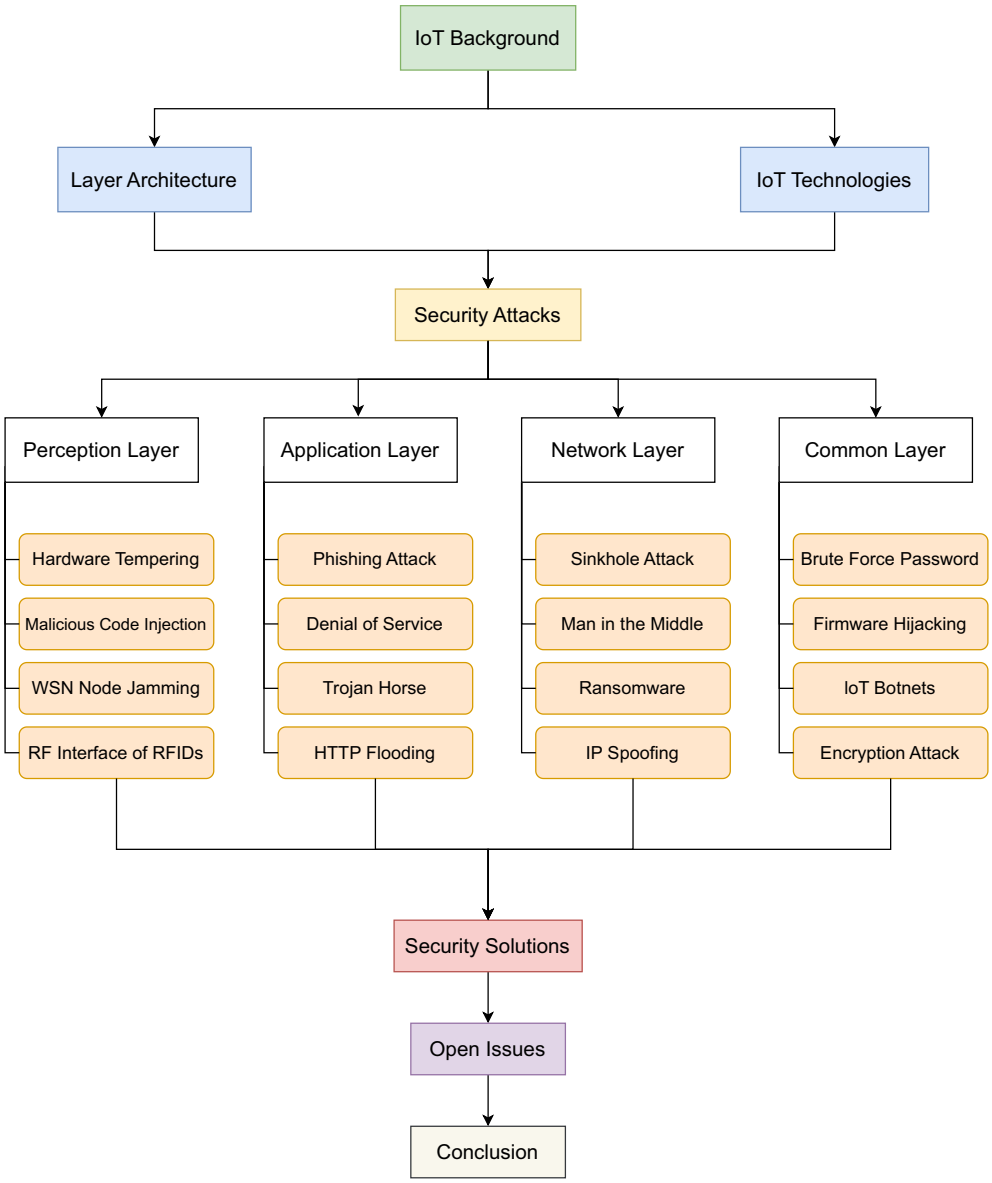

FIGURE 6 Sequence flow of the proposed survey.

1. Zigbee coordinator (ZC): This is the main and most capable device. The coordinator coordinates the whole network of communication. It helps in channel selection and provides an ID to the network. It provides a unique address to each device in communication. It also helps in transferring messages in the network. It also stores the information of the router.
2. Zigbee router (ZR): It is an intermediate node between the coordinator and end device. It receives data from the coordinator and sends data between different nodes. There may be more than one end device for one router.
3. Zigbee end device (ZED): The end devices have the information to talk to the router. Sometimes when there is no signal, the end devices sleep (standby mode); because of this, there is a saving in energy consumption which makes Zigbee more efficient. The end devices accept the different information from the parent node.[19]

Zigbee can be used in many devices, such as Smoke Detector Automation in home appliances like lights, fans, locks, and so forth. Wireless sensor network, Medical data collection, Industrial control system, Smart parking slots, Smart transportation system. Tarek Elarbi et al.[20] proposed a design of Zigbee transmitter for IoT devices. They also analyzed the design of the Cyclic Redundancy Check (CRC) and Bit to Symbol blocks of the ZigBee transmitter. And run the simulation process of CRC, Bit to Symbol blocks using Verilog HDL. Ana Koren et al.[21] presented an energy-efficient Zigbee Body

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Area Network (BAN) model to provide quality solutions for IoT-based Smart Homes in the fields of both BAN and health. BAN contains IoT devices wearable on human body parts. They also stimulated the efficiency of the model through the simulator. Due to massive development in smart devices, BAN and IoT are becoming major key solutions in reporting issues related to health and accidents. Fal Sadikin et al.[22] presented various methods to detect attacks on Zigbee-based IoT systems. They explained their Intrusion Detection System (IDS) implementation, which can detect different types of familiar attacks and potentially new types of attacks in Zigbee-based IoT systems with the help of machine learning-based anomaly detection.

Today, security has become a major issue that is slowly being solved. The utmost requirement of security is to protect the data from malicious persons. There are many security threats to Zigbee technology, such as:

1. Bus pirate: It is a type of Universal Bus developed for programming, debugging micro-controllers, Integrated Circuit (IC), and so forth. The newer version of Bus Pirates has the technology to enter the user computer through USBs and wired communication.[23]
2. GoodFet: It is the JTAG adapter, which is based upon TIMSP430FET UIF. This came after HackADay bus for considering it as a USB interface.
3. DOS attack: Denial Of Service (DOS) is the type of threat in which many dummy users send requests to any server, which may cause a temporary or permanent server breakage. Here in Zigbee, the dummy request may create disturbance in the IOT device.[24]

2.1.2 | RFID

Radio Frequency Identification (RFID) is mainly used for tracking objects, parameters, and so forth. This technology automatically identifies objects (such as items inside a mall, fastag, some cards, etc.) with RFID tags with the help of radio waves. It works on the principle of conversion of power to radio signals for communication. Here, this technology contains two main parts named: (i) RFID tag

Now, if we talk about the RFID tag, it is a sticker that must be installed on the object we want to trace. The RFID tag contains a transponder. The RFID tags can be read-only or read-write.[25] The RFID tag is divided into three categories:

1. Active tag: The tags used to have their power supply. Moreover, the power is sufficient to transport the feedback signal back to the reader.
2. Passive tag: This tag relies fully on the RFID reader. Here, the reader provides the energy to the tag to transfer signals. Because of this, they have a low range.
3. Semi-passive tag: This type of tag does not rely completely on the reader. They have their energy, but it is insufficient for them to give feedback. So, for feedback, they have to take energy from the reader.

The main benefit of this technology is that there is no requirement for line of sight. It means that the RFID reader machine should be in the range of the RFID reader. The tag contains various parts, such as:

1. Transponder: It receives and transports the radio waves from and to the reader.
2. Rectifier circuit: Here, the capacitor inside the rectifier circuit stores the energy for the transportation of radio waves inside the tag.
3. Controller: It uses the capacitor energy, giving the reader feedback.
4. Memory: It stores the information received through radio waves.

The RFID tags mainly work on three frequency ranges: Low Frequency (LF), which has a frequency of 125 to 135 KHz and a range of around 10 cm; High Frequency (HF), which has a frequency of 13.56 MHz and a range of around 1 m, Ultra High Frequency (UHF) which has a frequency of 860 to 960 MHz and range of around 10 to 15 m. The Low and High-Frequency tags work on the principle of Inductive Coupling, which is Near Field Coupling, while UHF works on the principle of Electromagnetic Coupling, which is Far Field Coupling.

RFID reader is a combination of antenna and transceiver. It can be a sized scanner or as large as the door in malls, and so forth. The reader is of two types based on motion: fixed reader and mobile reader. As the name suggests, fixed readers

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are stationary readers, such as doors in shopping malls, and so forth, while mobile readers are in motion, such as hand readers. The RFID reader contains various parts such as:

1. Radio frequency signal generator: The function of the RF Signal Generator is to generate the radio waves.
2. Radio frequency detector/receiver: The RF detector/receiver receives the radio waves from the tag.
3. Microcontroller: It is the main processing unit of an RFID tag. The microcontroller processes the information and commands the generator to generate the radio waves. In return, the receiver receives the signal and sends it to the microcontroller.

Thomas Hänel et al.[26] developed a security system that can increase security levels, mainly sensor modules based on RFID tags using Message Queuing Telemetry Transport (MQTT) protocol, which is most commonly used in IoT. Safkhani et al.[27] categorized modern ultra-lightweight mutual authentication protocols into two groups and showed their vulnerabilities to replay and desynchronize attacks. And proposed a more generalized improved mutual authentication protocol with a new message authentication code function and proved the security requirement for lightweight protocols.

There are many ways in which they attack RFID. Some of the security threats are mentioned here such as:

1. Clandestine: It means to work with something hidden. So, the attacker enters the RFID tag through malicious code and steals the data stored earlier.
2. Tracking: Here, the attacker tracks the functionality of the RFID devices, because of which the attacker can easily steal the information.
3. Skimming: Under this attack, the attacker wirelessly interferes and steals private information such as card details, etc., by storing the same in tiny microchips through magnets and then accessing it.[28]
4. Cloning: It means to create a duplicate of the original. Here, the attacker clones the original chips to another chip and adds some other controlling functionality in the cloned chips to get control over the RFID device.[29]

Some of the applications of RFID tags are Access control, Quality checking, Tracking objects, FASTag, managing supply chain, and so forth.[30]

2.1.3 | Bluetooth

Bluetooth is a short-range (around 10–100 m) radio communication that uses radio frequency to communicate between devices from the IoT network.[6] It is established on IEEE 802.15.1 standard. Bluetooth uses Frequency Hopping Spread Spectrum (FHSS) technology for all transmissions. Bluetooth establishes a Wireless Personal Area Network (WPAN) to provide a speed of 1 Mbps without the need to connect physically. Nowadays, Bluetooth technology is embedded already in mobile phones, laptops, headsets, wearable devices(i.e., smart watches, smart shoes), printers, radio devices, and so forth.[31] Bluetooth is also used in vehicle entertainment facilities, vehicle engine health monitoring, traffic management, bar code scanners, medical instruments, and so forth. To communicate between Bluetooth devices, the communication channel must be established by having the devices associated and authenticated. This process is called pairing. The updated version of Bluetooth is called Bluetooth Low Energy (BLE), also known as Bluetooth Smart. The difference between the older versions of Bluetooth and BLE is that BLE enables lower-power communication. With this, the network life expectancy increases significantly. Its range is also 10 times the older version of Bluetooth.

There are four security modes in Bluetooth. Security modes define when and where security procedures will be initiated. Security mode 1 does not initiate any security procedure. Security procedures are started in Security Mode 2, a service level-enforced security mode, after connection establishment but before channel establishment. All devices can be authenticated and encrypted using Bluetooth devices in Security Mode 3. Security procedure can be started after link setup with security mode 4. To provide privacy, Bluetooth additionally provides three different encryption options. In encryption mode 1, none of the connections are encrypted. Encryption mode 2: Individually addressed connections are encrypted using encryption keys obtained from individual link keys; broadcast connections are not encrypted in this mode. In encryption mode 3, a key based on the master link key is used to encrypt each connection.[4]

Which security modes should be incorporated in the manufactured gadgets is decided by the manufacturers. These devices then decide which devices should be included on the "trusted devices" list so they can communicate without asking for permission. The devices are shielded from unauthorized data transmission by these security settings.[18] Goh et al.[32]

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developed an Indoor Locating System (ILS) using BLE for IoT systems, which is a simple and low-cost implementation through which users can attach a Bluetooth tag to their belongings to get track of it and reduce their wastage of time. Katila et al.[33] discussed various potential mobility assistance strategies for IoT systems that use high-rate BLE. These systems are put into practice and tested in an office setting. Numerous experiments are used to examine the latency and power consumption findings of a sensor device. When larger data sets are transferred more quickly, it can save energy.

Like other wireless technologies, Bluetooth is susceptible to assaults. Attacks of all kinds, including man-in-the-middle (MITM), DOS, eavesdropping, message manipulation, and so forth, can be taken into account. Here are some attacks that should be considered:

1. Bluejacking: It involves SMS transmission to adjacent Bluetooth devices. Bluejacker can be saved as a contact if the attack is not adequately documented. After adding the contact, they can submit content that will be opened immediately because it comes from a known source.
2. Car Whisperer: Hackers get access to a car entertainment system and transmit or receive data through the microphone or speaker.
3. Bluesnarfing: Using a Bluetooth connection to steal data from a device is known as Bluesnarfing. Contact lists, emails, text messages, and occasionally photographs or videos are all examples of the information that can be used.[34]

2.1.4 | WiFi

Wireless Fidelity is referred to as WiFi. It follows the IEEE 802.11 standard. It uses 2.4 and 5 GHz frequencies to construct Wireless Local Area Network (WLAN). It can communicate at a distance of 100 m with a maximum speed of about 100 Mbps.[31] It provides Internet as a radio signal to the devices connected wirelessly to the WiFi. WiFi provides security to the attacks that target its vulnerability to not detect the threats in the network. To provide encryption, there are four encryption levels in the WiFi: WEP-64 bit encryption, WEP-128 bit encryption, WPA-256 bit encryption, WPA2-PSK (TKIP) and WPA2-PSK (AES)

These are encryption levels that protect WiFi from potential threats. WEP stands for Wired Equivalent Privacy. WPA implies WiFi Protected Access. Most WiFi networks support WEP 128-bit encryption. But it is not strong enough to fend off the attacks. So, WPA and WPA2 are used as substitutes. WPA enables 256-bit encryption, while WPA2 supports TKIP or AES encryption, the most secure encryption algorithm. For a secure WiFi connection, WPA2 with AES encryption is advised. WiFi provides security levels to protect the user in the networks. The working of the security level can be explained in the following steps:

1. Security mode 1: Information about WiFi networks is analyzed generally.
2. Security mode 2: Enables WEP, WPA, and WPA2 encryption.
3. Security mode 3: It uses MAC address filtering to block unauthorized devices.
4. Security mode 4: It allows the required protocols to be activated in the communication network.
5. Security mode 5: Disables Service Set Identifier (SSID) transmission across the communication network. When other users try to see the accessible wireless networks in your vicinity, they won't be able to see your wireless network name because you've done this.
6. Security mode 6: Assigns IP address. It can be either static or dynamic. Static IP address increases security.[35]

Agyemang et al.[36] demonstrated a lightweight technique that ensures WiFi-enabled IoT devices can discriminate between legal and illegitimate access points. Additionally, its effectiveness and performance were assessed. Alyami et al.[37] showed that fingerprinting devices of IoT systems are possible without joining the WiFi network of IoT devices. And also they discussed potential defenses to mitigate the attack.

Even with these precautions, as with other types of networks, WiFi is still vulnerable to attacks. The threats to the WiFi networks are informed below:

1. Wireless network eavesdropping: Gathering data from snooping the network is called eavesdropping. Wireless Network Eavesdropping is easier to achieve than wired network eavesdropping. To prevent eavesdropping, we can encrypt the data. But still, data will be eavesdropped in its encrypted form. We can modify the WiFi network to secure our network.[37]

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2. Search wireless signal attack: If the WiFi network is not properly encrypted, then the devices searching them might be attacked by hackers. To avoid that, we can encrypt the WiFi networks, make SSID announcing disabled, and so forth.
3. Lack of encryption: Basic security applications use WEP encryption. WEP encryption is a weak security encryption to the standard. So hackers take advantage of the WEP encryption and attack the network. That why, for better security, the WPA2 AES encryption algorithm is recommended for use in WiFi networks.

2.1.5 | Ethernet

Ethernet is used for Local Area Network (LAN) to connect the devices to the internet. It works on IEEE 802.3 standards using physical and data link layers of TCP/IP protocol to tell the devices how to share their data with other devices through physical links. Ethernet LAN technology connects the computer to the router to access the Internet, using Ethernet cable as a physical medium. Various types of Ethernet cables are available in the market per the requirements. The transmitted data speed also depends on the Ethernet cable type and can be limited by the cable administrator. The most commonly used Ethernet cable is co-axial. Other types of cables are twisted pair, fiber optic, and so forth. Some standard co-axial cables are as follows:

1. 10BASE-T: This is the IEEE standard name of an Ethernet cable. 10 represents the transmission speed of data transfer in Mbps. BASE refers to the type of signal which are being transmitted. And T refers to the type of the cable. As explained, this cable can send data at a maximum speed of 10 Mbps baseband signal in the twisted pair cable.
2. 100BASE-T: It sends data at the speed of 100 Mbps.
3. 1000BASE-T: It sends data at the speed of 1000 Mbps. Sometimes it is called Gigabit Ethernet.[38]

Nowadays, the commonly used Ethernet cable can send data at the speed of 1 GBPS with a maximum bandwidth of 100 MHz. Ethernet network is mainly used for short-range transmission of data. It needs to be optimized for long-range data transmission. And it also has advantages over the wireless connection of devices. It is reliable for data transmission and uses a common firewall for data security. However, it could be more perfectly safe as various types of attacks can threaten the devices of the ethernet network. Some are described below:

1. Unauthorized joins: It is a physical type of attack. An attacker can use an unconnected or another computer port to connect an unauthorized device and harm the network.
2. MITM attack: MITM attack allows the attacker to hear the conversation between two legitimate devices, which should be secure. Its most occurrences can be at public WiFi where you are vulnerable to the open network shared by many.
3. Session hijacking: To avoid needing the user to write their password on each page, most web applications utilize a login technique that creates a temporary session token for subsequent requests. An attacker may attempt to discover a user session token and use it to send requests on the victim behalf. Once the attacker obtains a session token, spoofing is not necessary.[39]

2.1.6 | Augmented reality

Augmented reality (AR) is the new emerging technology in the modern world. It is a way of interacting with computers. It works on implanting digital objects in the real world through some algorithms that measure distance, surface, and so forth. They usually use a camera, but this can sometimes be achieved through Depth lenses. It maps the surrounding environment and tracks the location for keeping virtual objects. The camera present on the device recognizes the objects and sends the information to the cloud. The device inside the AR glasses downloads the information from the cloud for further processing, like we load the website page from the Universal Resource Locator. When the user moves, the distance between the object and the glass will change, which is sensed by the lens, and according to that, the size of the virtual object will also change. There is a 3D Digital twin inside the cloud. This twin acts as the overpass between the virtual object and AR. The twin collects the information from the product, business system, and so forth, to know the current details of that product or system.[40] The following components can achieve AR:

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1. Depth sensing camera: This type of sensing camera captures the exact distance, location, and angle of the object concerning the base level.
2. Registration tools: They locate the space in memory for storing the information regarding the object.
3. Computer vision: It stores the image for future reference and analyzes the pixels of the image through a Machine Learning algorithm. So whenever the same object type comes under the lens in the future, it would recognize the pixels and give the output according to the machine learning algorithm.
4. Output device: This can be termed as the screen or device in which the final output of the process is being shown.

Today, AR is being widely used in many fields, such as:

1. AR in smartphones: Some smartphone applications use satellites to collect information. They store all the essential information regarding that place and display it on a mobile screen when the object is nearby. Moreover, some medical applications help the doctor identify the type of physical wound through machine learning algorithms. Further, some games also work on AR technology. They connect your real world with some virtual environment.
2. Vision-based people tracking: This application allows us to track people. Here, when the people enter the defined range of the camera, they will be given an individual particulate filter for the system. Now, this filter will be continuously tracked, and as new people enter, they will be given different particulate filters to the system. When the person reaches out of the range of the camera, that filter is automatically deleted.[41]
3. AR in military: In the military, this was being used for many years. They are using this technology to teach the freshers about different types of guns, their range, weight, dimensions, and so forth. Moreover, this can also be used as the simulator to practice shooting through virtual guns as they are safe.[42]

Gandhi et al.[43] discussed Distributed Denial-Of-Service (DDoS) attacks on IoT, cloud, and AR. Liu et al.[44] analyzed the effects of AR on physical education. They suggested an AR solution for school physical education training based on a cloud network, IoT, and remote users. This technology is beneficial, but on the other hand, there are some security threats to this technology, such as:

1. Malware: In this attack, malware is malicious software that the user may unknowingly install to perform different illegal actions without the administrator permission. Because of this attack, sometimes it may show some wrong information and may damage the operating system
2. Physical damage: This is a type of attack in which the AR will not work perfectly if the internal chip breaks for some reason.
3. Deep fakes: It can be done through fake photos, videos, images, and so forth. Here, there are fake photos based on AI-manipulated fabrication, which can be pasted on another photo to change the identity or to spread the wrong message.[45]
4. Health effect: This technology has proved to be addictive sometimes to some people. Moreover, it will decrease a person reactive time.
5. Privacy: Using this technology is somewhat risky because if the hacker enters the device, then it can easily track your activity, and sometimes, there are chances that the hacker may blackmail you for not disclosing some personal information.

2.1.7 | E-SIM

The embedded subscriber identification module, which is also known as E-Sim, is a technology that requires no physical SIM card. This can already be installed in some devices, so the user can access this technology by activating it once.[46] This technology is also named UICC, a Universal IC Card. The main benefit of this technology is that if the user wants to change the service provider, then the user has to install a new card to replace the old card. As there is no SIM card in eSim, the user can easily change the service provider without changing anything. The Embedded SIM technology can considered to be the better technology as the safety will increase compared to the recent scenario. There are many applications of this E-SIM, such as:

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1. E-SIM in vehicle tracking: This technology is more beneficial for the trucks because the truck owners constantly monitor the movement through mobile. The advantage of this technology in vehicle tracking over any other is that as the truck travel through the various region, the can flawlessly change the network carrier without any extra charges.
2. Stationery monitoring device in brutal conditions: Sometimes, there are areas where it is very difficult or rarely impossible to implant IoT devices, so we use this technology. The E-SIM technology comes with waterproof and vacuum-packed. So they can easily implant in water, mud, snow, and so forth. In this way, they help in keeping track of that area.
3. Connected car: There is a fitted inside the car that collects all the information inside and outside the car and sends it to the user interface through a WLAN as it supports HF for data sending through frequency hopping or direct sequence.[47]

This technology is beneficial in terms of privacy, but some security threats belong to E-SIM. They are as follows:

1. SIM jacking: Under this attack, the attacker accesses the person personal information. Then, he may contact the telecom provider and mimic you. They tell the company that their device had been stolen so that the company could provide a new SIM card. After some security checking, the company may provide a new SIM card with the same number. So now the attacker has full access to the SIM as he can handle and respond to the messages or calls. Moreover, he can access the personal data stored from that mobile number. In this way, the attacker attacks through E-SIM.[48]
2. Identity fraud: It means gaining control over another user information for fraud. Here, the scammer gains control over the user personal information by changing his original identity to access various bank accounts, financial transactions, and so forth. Individuals and businesses may suffer a lot due to this type of attack.
3. Device poisoning: This kind of attack allows for the tempering of user data, which can be utilized as training data to get better results. As a result, the original information may alter, which causes a barrier for the user.

2.1.8 | Smart sensors

Today, the smart sensors have become the part of life. Most of the things near our environment are loaded with smart sensors. The intelligent sensor consists of an analog or digital transducer merged with sensing and evaluating abilities. It contains transduction, signal conditioning, and a processor with well defined algorithms. They are used for converting analog to digital, information processing, and so forth. The main advantages of using smart sensors are low cost, minute size, low energy consumption, and so forth. These smart sensors are used in a wide range of technologies, which are as follows:

1. Smart dustbin: There is a smart dustbin concept with sensors in it. So when the dustbin is 90% full, the deployed sensor will generate the message to squeeze the waste to increase the dustbin capacity. At last, when the dustbin is full, it sends a message to the nearby garbage collector.
2. Smart lights: Intelligent sensors work on the principle of light modulation. The sensor inside the smart light senses the daylight; when it detects the light intensity below some candela, it automatically sends the signal to turn on the lights. Similarly, it works for turning off the lights when the intensity is above some defined candela.
3. Smart healthcare devices: Today, many innovations are taking place in medical science. So, all modern health-related devices like blood pressure monitors, oxymeters, sonography machines, CT scan machines, cardiogram machines, and so forth, use different sensors. The sensor collects the data from various parts of the human body, processes the information according to the defined algorithm, and finally displays the output.
4. Smart cars: These cars are termed as the future robots. Some companies manufacture automatic or driverless vehicles. The sensor attached to the front of the car senses the distance between the car and the front obstacle and automatically applies the brake or changes the lane. There are rear-view mirror sensors that automatically get dim when the light intensity is reduced. Moreover, there are rear parking sensors that work similarly.

These sensors are termed to be an essential part of our lives, but there are some security threats to these sensors, such as:

1. Information leakage: In this attack, the unidentified user steals information, such as passwords, timings, data, and so forth, from the sensors.

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2. Transmitting malicious sensor command: When inside any sensor, if any malicious pattern code activates, the sensor may generate unexpected output and communicate to an unknown source for sharing information.
3. False sensor data injection: This attack can be explained as the attack in which some false data is forcefully entered inside the sensor. Then, the sensor will work according to the false data input and lose its original functionality.
4. DOS: It is an attack in which the sensors are forced not to work. This threat attack on one of the sensors may lead to dysfunctionality in all other devices connected with that device.[49]

Table 2 shows the comparative analysis of various IoT technologies along with their possible security threats that can compromise the data and Table 3 shows a comparative analysis of numerous state-of-the-art IoT technologies.

2.2 | IoT-layered architecture

Figure 7 depicts the three-layered IoT architecture, which is bifurcated into the Perception Layer, Network Layer, and Application Layer.

TABLE 2 Comparative analysis of various IoT technologies along with their security threats.

Technology	Scope	Pros	Cons	Security threats
Zigbee	Communication protocol for short range	For low cost & high reliability	As it is low bit, Transmission rate is slow	Bus Pirate, GoodFet, DOS attack
Bluetooth	Short range communication using UHF radio waves	Communication without wire or cable Easy and Affordable and Efficient	DOS, Low range communication, Eavesdropping	Bluejacking, Car Whisperer, Bluesnarfing
RFID	Tracking and Trasmission of various factors	Speed, Convenience, Diverse	Problem in Scanning water and metal, High cost, Low Security	Clandestine Scanning, Tracking, Skimming, Cloning
Wifi	Essential IOT enabler	Long range Communication, Provides Wireless connection	Frequency issues, Limited bandwidth, Health issues	Wireless network eavesdropping, search wireless signal attack, lack of encryption, Malware
Ethernet	Traditional tech. for fixed & wired communication	High speed and reliability, relatively inexpensive, data quality s good	Limited flexibility, Physical Vulnerability, Not suitable for high traffic	Unauthorized joins, remote access to LAN, MITM, ARP poisoning, Session hijacking
Augmented reality	Putting existing world into virtual environment	Wonderful experience, help to create virtual world	Affordable, Security issues, Health Issues, Addictive, sometime risky	Malware, Physical damage, Deepfakes, Health issues, Privacy
eSIM	A SIM which would carry information digitally via air	Easy switching networks, no need to keep sim physically	Backup taking is hard as no sim is there, Less availability	SIM jacking, Identity fraud, device poisoning
Smart sensor	A device which takes input and perform diff. algorithm	Sense all material, minute size, Long life	Packaging and handling is tedious task, sometimes slow response	Information leakage, Transferring infected sensor command, False sensor data injection, DOS

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TABLE 3 Comparative analysis of numerous state-of-the-art IoT technologies.

Technology	Author	Objective	Pros	Cons
Zigbee	Elarabi et al.[20]	Design and simulation of Zigbee transmitter for IoT devices	Digital design of Zigbee data transmitter will allow accurate data transmission with low power requirements	The Symbol-to-Chip block (where DSSS technique is performed) is not developed.
	Koren et al.[21]	Model a Zigbee network in IoT based Smart Homes	Proposed a feasible energy efficient model for BAN in IoT network of Smart Homes	MAC protocols mentioned in the study based on schedule contention and LPL are unable to accommodate on-demand and emergency traffic.
	Sadikin et al.[22]	A ZigBee Intrusion Detection for IoT based systems	It provides anomaly detection in addition with accurate detecition from known attacks	It may give false alarm in real IDS deployments
RFID	Hänel et al.[26]	Adjustable security system for RFID equipped IoT devices	It removes the need for pre-equipping the devices with security information	Attacks like getting the root access of devices are not discussed in the paper
	Safkhani et al.[27]	Improving RFID- IoT based generalized protocols	Proposed a improved mutual authentication protocol (GIMAP)	Application specific features are not shown in this study
Bluetooth	Goh et al.[32]	IoT based indoor locating system using BLE	It represents simple, practical and low cost system using BLE to develop ILS for IoT devices	The system has managed to obtain a maximum average accuracy of 69.584%
	Katila et al.[33]	Analysis of mobility support approaches for IoT systems using BLE 5	The presented approaches can reduce energy consumption while transferring larger data sizes at faster speed	Presented cases are not sufficient for real-time application in specific condition
WiFi	Agyemang et al.[36]	Rogue access point detection algorithm for WIFi enabled IoT devices	It represents real-time and lightweight intrusion detection algorithm.	The algorithm doesn't detect MITM attacks at gateway of IoT networks.
	Alyami et al.[37]	Attack scenario of eavesdropping on encrypted WiFi based IoT devices	Out-of-the-network user is considered as an attacker and demonstrated information leakage in IoT network	This method do not investigate on additional modes and fingerprinting of app
AR	Gandhi et al.[43]	Security and privacy issues in IoT, cloud and AR	Various DDoS attack are discussed for IoT, cloud computing and AR	A detailed description of related attacks are not given
	Liu et al.[44]	AR technology for school physical education	AR training method is efficient and in sports and physical education	The operating environment is costly and highly complex

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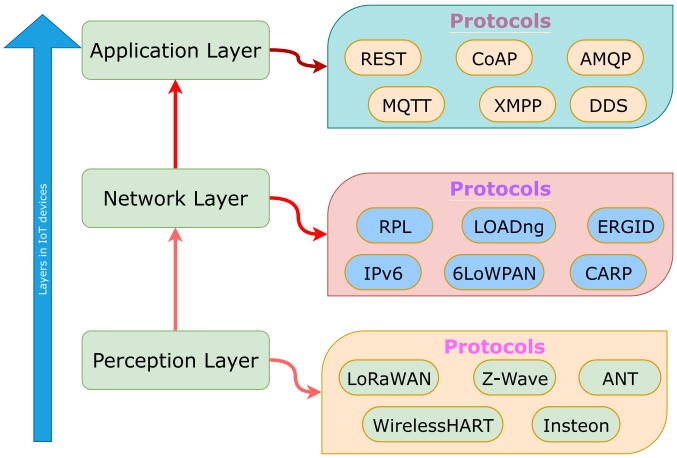

FIGURE 7 IoT-layered architecture.

2.2.1 | Perception layer

The perception layer, or the Physical layer, works at the lowermost level in a layered architecture. The main work of this layer is to measure, collect, and sense the collected information. Hence, this layer is also known as the Sense layer in IoT. This layer is interconnected with many of the physical device components.[13] The protocol in the layer helps to make the communication process easy and efficient. There are various protocols affiliated with this layer. They are as follows:

1. **LoRaWAN**–Low power wide area networking (LPWAN) is generally used for radio modulation. It is the Media Access Control Protocol of this layer. This protocol connects to the various types of IOT devices wirelessly between the end user device and different gateways. This protocol was developed for battery-operated and wireless devices*. The device can be connected with the help of LoRaWAN to network by two methods:
   - Over the air activation (OTAA): For communication, we need to connect the network layer ID with the application session ID to establish a link between sender and receiver.
   - Activation by personalization (ABP): To make the communication process easier, some devices have hard-coded keys, but this is less secure.
2. **Z Waves**: It generally used for Home Automation Networking (HAN). HAN includes smart lights, coolers, doors, curtains, and so forth. This protocol works on low power to increase efficiency. This protocol was first developed by ZenSys (currently Sigma device). The upgraded version of Z waves, developed by Z Wave Alliance, ranges nearly 30 meters in point-to-point connection†. As the range increased, the data transferring capacity of the Z waves was still low. So, it is used in devices with a low data requirement, such as turning lights on or off, fire detection alarms, maintaining room temperature, and so forth. Z waves work on an ISM band of nearly 900 MHz. The latest versions of Z waves have a transmission speed of nearly 200 Kbps. There are various features of this protocol. They are as follows:
   - The installation process in various applications is easy for this protocol.
   - As this protocol sends a small amount of data, it uses low power.
   - It also supports Interoperability. This means different systems, technology, devices, and so forth, connect and communicate in a coordinated way without the help of the user.
3. **Wireless-HART**: It is a layered protocol arranged like a stack and open to the public. This protocol was developed by HART Communication and Foundation Company in the year 2004. It works on the free licensed radio waves of frequency 2.4 GHz and complies with IEEE 802.15.4-2006‡. This protocol is generally used for automation in Industrial areas. The licensed data rate is 250 kbps, and the range is 225 m for this protocol. This protocol has various features, which are as follows:

*Carvalho Silva dJ, Rodrigues JJPC, Alberti AM, Solic P, Aquino ALL. LoRaWAN—A low power WAN protocol for Internet of Things: A review and opportunities. In: 2017: 1–6.

†Krejčí R, Hujňák O, Švepeš M. Security survey of the IoT wireless protocols. In: 2017: 1–4.

‡Liao JM, Ge JL. Study on wireless HART network layer. In: 2010: 187–18.

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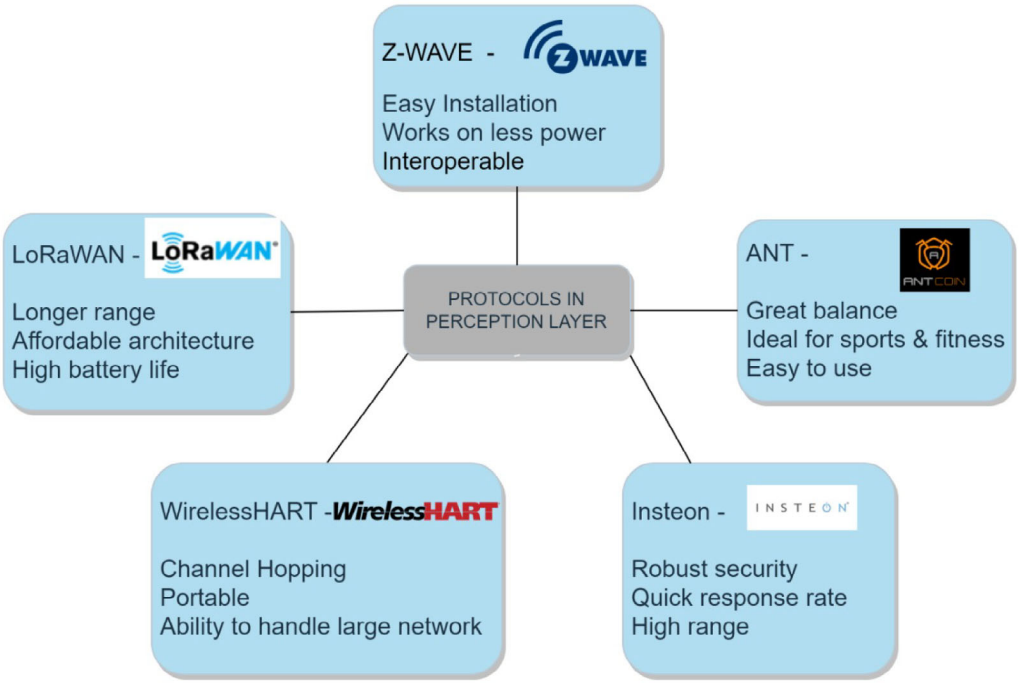

FIGURE 8 Protocols in perception layer.

- It also follows channel hopping, which means changing the channel for the communication process, which may increase security.
- This protocol is immune to handling large networks. It can also accept many requests at the same time.

4. **Insteon**: This protocol was developed by the Insteon company. This was made to communicate with home devices with the help of RF, powerline, and both. There is a hub which has to be connected to the internet. All the other home automation devices send the data through this protocol, which is further processed by the hub. There are some features associated with it:
   - This protocol provides robust security, which means that it ensures secure communication.
   - This protocol has a quick response rate as it can handle requests too quickly.
   - It also supports high range, or we can say that it has long-range coverability.
5. **ANT**: The ANT protocol is 2.4 GHz wireless radio communication protocol. This protocol helps in sending information wirelessly from one device to another device. This protocol has been proven as the ultra-low power wireless communication protocol. The ANT+ provides an ecosystem for devices built on Ant and thus provides efficient communication. This protocol provides great balance, which means that it provides power to the sensor and display in a balanced form, which helps reduce power consumption. It can be considered ideal for sports and fitness as these sensors are proven to be ultra-low power wireless protocols. This protocol is simple to use, and it has very user-friendly interfaces[§]. Figure 8 shows the communication protocols and their features discussed for the perception layer.

2.2.2 | Network layer

It is implemented between the Application layer and the Perception layer. As soon as it receives the data from the perception layer, it sends it to the appropriate application or devices at the application layer. It covers transmission over short distances or in isolated locations.[7] The short-distance transmission is achieved through wired or wireless technologies like WiFi, Bluetooth, LTE, Zigbee, Ad-hoc, Mesh, and so forth.[13] Satellite communication networks, mobile communication networks, hubs, gateway, cloud computing, routers, and so forth, accomplish remote transmission. The network

[§]Mehmood NQ, Culmone R. An ANT+ Protocol Based Health Care System. In: 2015: 193–198.

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layer is essential because it transmits the data from the perception layer to the application layer reliably and securely with the help of numerous protocols and technologies. We have discussed some of them as follows:

1. **RPL**: The Routing Protocol for Low Power (RPL) is used in networks with limited resources. It is an IEEE 802.15.4-based vector-based protocol. It is appropriate for multi-to-one communication, as well as one-on-one communication, and it supports several Data-Link protocols. The Destination-Oriented Directed Acyclic Graph (DODAG) produced by RPL has just one path from each leaf node to the root node. Each node broadcasts its status as the root node using a DODAG Information Object. The network transmission of this data gradually produces a complete DODAG. When a new node requests to join the network, the root node responds with a DAO acknowledgment, allowing the new node to do so[¶].
2. **LOADng**: The Lightweight On-Demand Ad Hoc Distance-Vector Routing Protocol-Next Generation (LOADng) is an improved version of the Ad-hoc On-Demand Distant Vector (AODV) with the properties of a lightweight and responsive protocol. It finds the path both to the source and to the destination. If the routing procedure fails, it tries to rectify it. If not, it informs the appropriate node of the failure[#].
3. **CARP**: Channel-Aware Routing Protocol (CARP) is a routing protocol developed for underwater communication because of its lightweight configuration. The link quality is considered when choosing the forward node for transmission. Data forwarding and network startup are the two procedures it carries out. Since the CARP protocol does not keep track of earlier data, it is of little use in situations where data is constantly changing. Therefore, E-CARP, an enhanced version of CARP that did away with CARP limitations, is used. The node can save previously received data thanks to E-CARP[‖].
4. **ERGID**: The Efficient Routing Protocol for Emergency Response Internet (ERGID) enables secure data exchange and emergency response in IoT applications. ERGID selects the optimum routing path to the communication destination based on the competing path nodes. Additionally, it ensures that the emergency service applications deliver real-time material and continuously enhances routing tables.
5. **IPv6**: The number of IoT device users in the database is growing exponentially. Due to the restricted number of permutations, IPv4 cannot accommodate all the devices with unique IP addresses. An improved version of IPv4 is IPv6. IPv4 addresses are 32 bits, but IPv6 addresses are 128 bits. Each of its eight 16-bit blocks, separated by semicolons, can be written as a 4-digit hexadecimal integer. A higher level of network complexity is made possible by IPv6 capability for auto-configuration, integrated security, and a range of additional mobility capabilities. IPv6 also has the advantage of being very effective at multicast communication, which ends the practice of broadcasting messages.
6. **6LoWPAN**: The IEEE 802.15.4 specifications for tiny packets are used by the IPv6 Low Power Personal Area Network (6LoWPAN). This lightweight protocol uses IP to transmit data over networks with low data rates. It is mostly utilized in homes and buildings for automation due to its limits in data transfer. Every device on the mesh network has an IPv6 address, allowing it to connect to the Internet instantly. It transmits data at a speed of 250 Kbps via a 2.4 GHz frequency spectrum to connect to the internet.

Figure 9 shows the visualization of the communication protocols highlighted for the network layer while discussing the IoT-layered architecture.

2.2.3 | Application layer

This layer interacts with the user to provide various services based on the processed information sent by the Network and Perception layers. It is implemented at the top of the architecture. It provides security and services using applications like intelligent health, smart transportation, smart homes, and so forth.[50] It is also the most vulnerable layer for many security attacks because of the lack of information of the user to fully utilize the characteristics of the IoT devices like

[¶]Granjal J, Monteiro E, Sá Silva J. Security for the Internet of Things: A Survey of Existing Protocols and Open Research Issues. IEEE Communications Surveys and Tutorials 2015; 17(3): 1294–1312. doi: 10.1109/COMST.2015.2388550.

[#]Bang AO, Rao UP, Visconti A, Brighente A, Conti M. An IoT Inventory Before Deployment: A Survey on IoT Protocols, Communication Technologies, Vulnerabilities, Attacks, and Future Research Directions. Computers and Security 2022; 123: 102914.

[‖]Gerodimos A, Maglaras L, Ferrag MA, Ayres N, Kantzavelou I. IoT: Communication protocols and security threats. Internet of Things and Cyber-Physical Systems 2023; 3: 1–13. doi: https://doi.org/10.1016/j.iotcps.2022.12.003.

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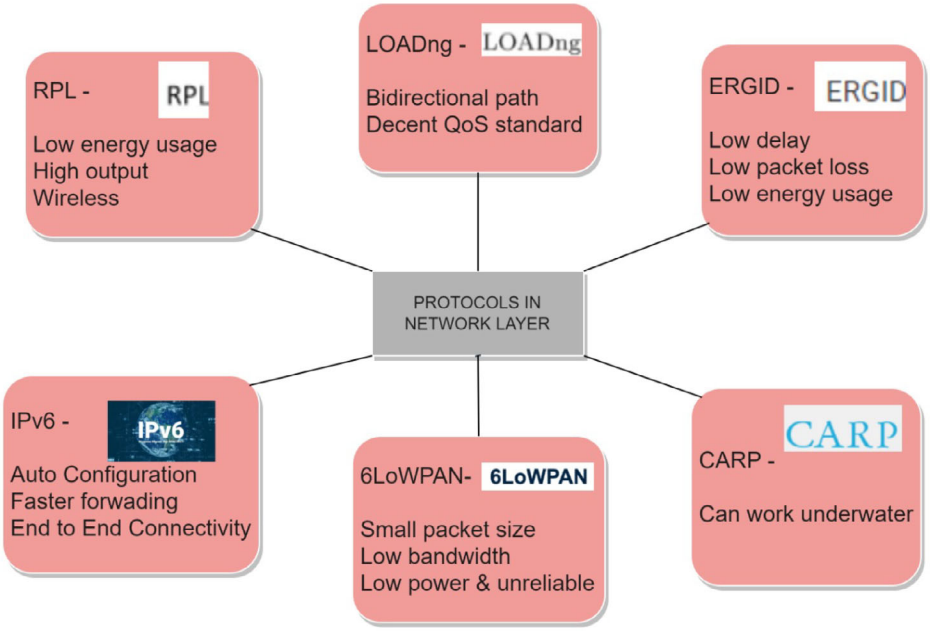

FIGURE 9 Protocols in network layer.

Confidentiality, Authentication, and so forth.[51] So this layer helps the user with Data integrity, Data confidentiality, Data authenticity. To provide solutions to the challenges of the technology revolution, we need application layer protocols with continuous upgradation to reach the end users' requirements. There are many protocols at the Application layer to support the functionality of the layer. Some of them are discussed below:

1. **REST**: Representational State Transfer (REST) works with web resources by their unique URL and delivers them to the user as HTTP, XML, or JSON files. So, it can be implemented in machine-to-machine and various devices[**]. The inconvenient part of the REST protocol is it is difficult to implement.
2. **CoAP**: Constrained Application Protocol (CoAP) is developed by the Internet Engineering Task Force (IETF) Constrained RESTful Environment (CoRE) Workgroup. CoAP is specifically helpful to the hardware which has less technological support. Devices that cannot use HTTP or TCP/IP can use CoAP. Because of the low power requirements of the UDP over TCP, CoAP uses UDP and IP protocol. It is a lightweight protocol for battery-powered IoT devices. It is a Request-Response Protocol that exchanges data asynchronously[††].
3. **XMPP**: Extensible Messages and Presence Protocol (XMPP) is designed to support both Request-Response and Publish-Subscribe models protocols. It is helpful in real-time communication to instantly communicate messages using end-to-end encryption and authentication. It is based on Extensible Markup Language (XML) to provide instant messages, multi-user chatting, video call meetings, and so forth. It is a language for encoding that is readable by both humans and computers[‡‡].
4. **MQTT**: MQTT is a Publish-Subscribe protocol designed by IBM. It is the most commonly used lightweight protocol in communication between machine and machine. This protocol is based on TCP protocol and can communicate in requirements networks like limited bandwidth, low battery resources, and so forth[§§].
5. **AMQP**: Advanced Message Queuing Protocol (AMQP) is a Publish-Subscribe model protocol. It mostly uses TCP protocols to send the message to other devices over the internet. It is mainly a point-to-point connection for connecting devices to exchange data[¶¶].

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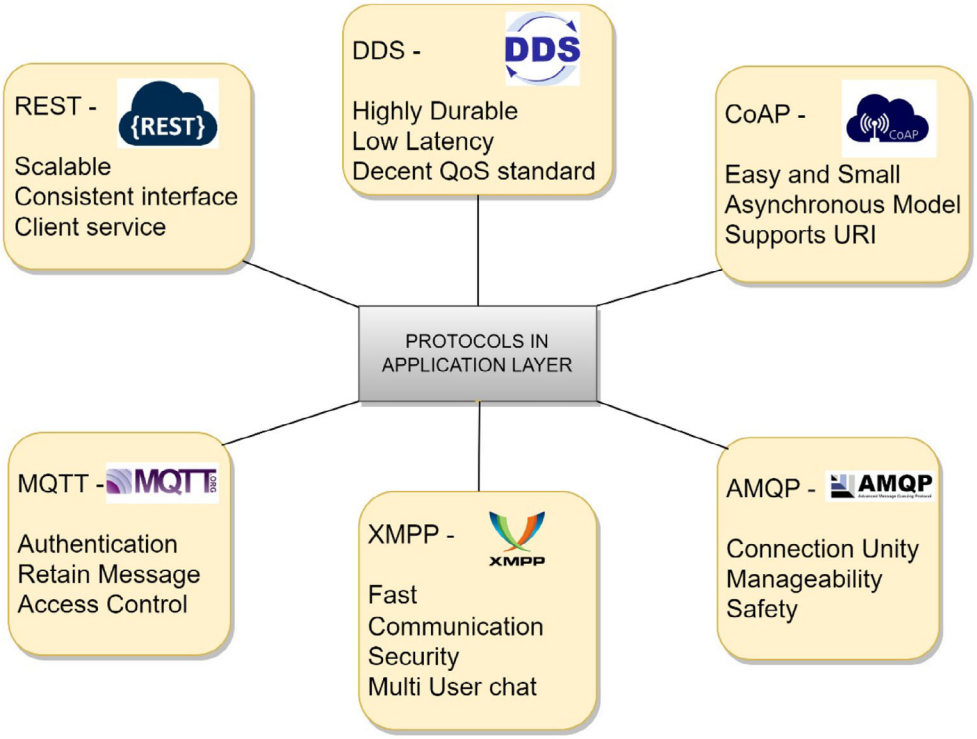

FIGURE 10 Protocols in application layer.

6. **DDS**: Data Distribution Service (DDS) is a Publish-Subscribe protocol that is created by Object Management Group (OMG). It provides Machine communication with broker-less architecture that can support the services to the link with security, reliability, durability, and so forth.

Figure 10 shows the communication protocols discussed for the application layer of the IoT-layered architecture of the comprehensive proposed survey.

3 | TAXONOMY OF IOT SECURITY ATTACKS

In the modern world, security is an essential requirement in any field. The term attack in IoT means forcefully accessing the information without the user permission. So, to get protection from this kind of attack, we require security, which is being provided against the attack in IoT. Figure 11 depicts the taxonomy of IoT security attacks corresponding to the different layers, such as the perception, network, and application layers. Furthermore, Table 4 shows the comparative analysis of security attacks discussed for the layers. Table 5 shows the abbreviations along with their definitions used in the proposed survey. There are various kinds of attacks which are associated with the various layer of IoT, which are as follows:

3.1 | Perception layer

This layer is executed at the bottom level in communication levels. This layer fundamentally gathers data from the censors and processes the data. There are various attacks associated with this layer, which are as follows:

1. **Hardware tempering**: The type of physical attack in which the hacker changes some hardware parts. The hacker modifies some circuits inside IoT devices to access the information. The information can be a communication key, routing table, cryptography key, radio key, and so forth.

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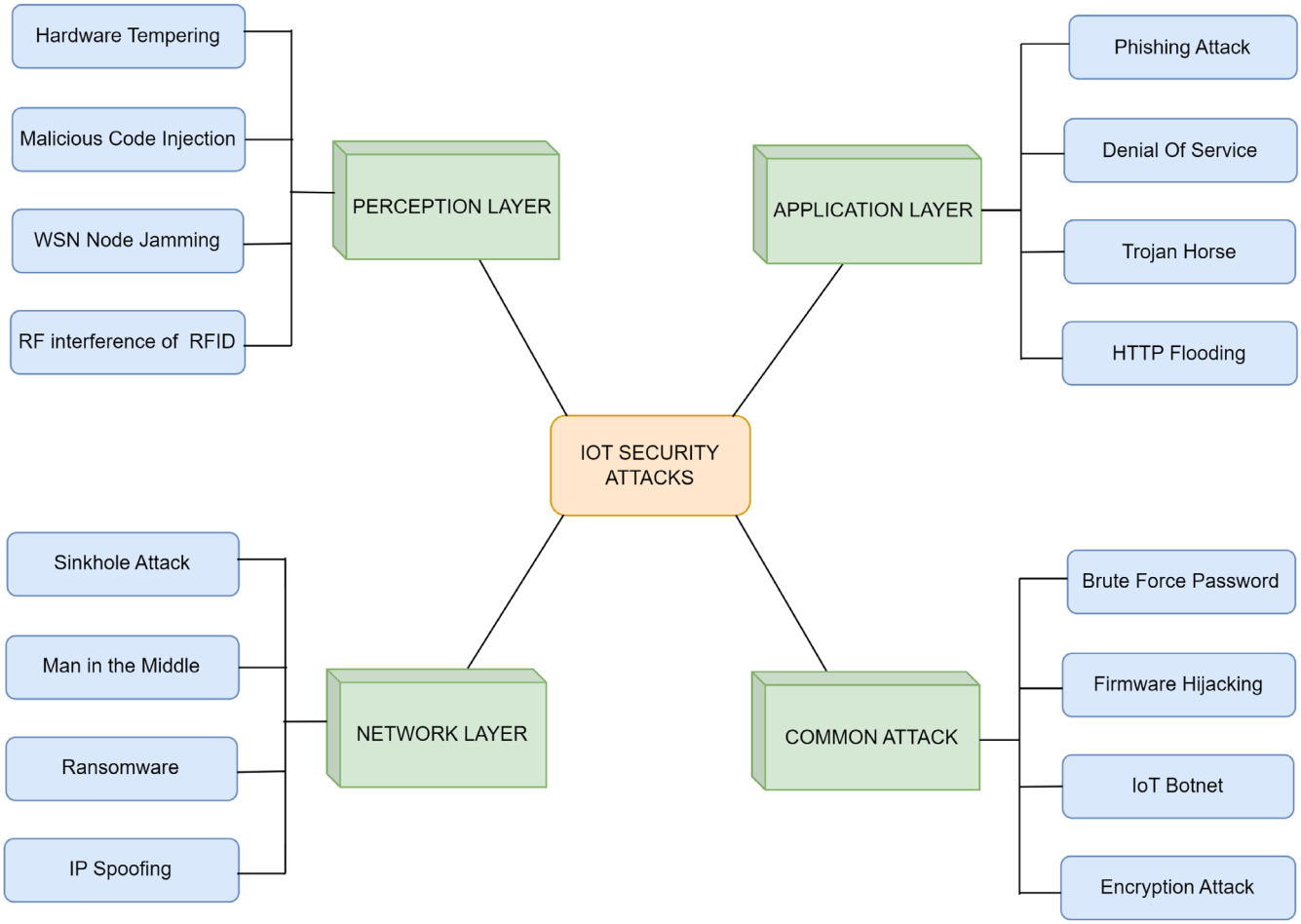

FIGURE 11 Taxonomy on IoT security attacks.

2. **Malicious code injection**: Some malicious code in the node is injected by the attacker. Due to this, the information is stolen by the hacker through the network. As a result, the network may be unavailable for some time.
3. **WSN node jamming**: Wireless Sensor Network(WSN) is a communication medium which works on radio frequency. If the packets transmitting through the channel exceed the network maximum capacity, then the network will jam, and the hacker uses the same concept. He sends a large number of dummy packets which carry no information. These dummy packets may jam the network. Shoukat Ali et al.[52] proposed different schemes like multi-hop, hierarchical, trust-based, and so forth. This method helps in protecting against the WSN node jamming black hole attack.
4. **RF interference of RFIDs**: As we discussed above, RFID uses radio frequency and helps track objects. So, in this attack, the hacker forcefully sends some noise through radio frequency. So, both radio frequencies interfere, and the original information may get lost. So, in this way, this noise signal becomes an obstacle in communication.

3.2 | Application layer

This layer is at the topmost level of communication. It acts as the interface for the user. This layer collects and shows the information or action that must be shown. There are some attacks applicable to this layer, which are as follows:

1. **Phishing attack**: This is the most ordinary attack. This attack is carried out first by generating a malicious link. These links are generated so that, at first glance, the links look like the original. Through this link, the hacker asks the user for some personal information recognized by the hacker.[53] Gopal et al.[54] proposed a Deep learning-based Artificial Neural

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TABLE 4 Comparative analysis of security attacks based on the layers.

Attack	Layer	Targets	Description	Counter measures
Hardware Tempering	Perception	Hardwares of Devices	Data Leak	Secure Hardware design
Malicious Code Injection		Hardwares of Network connection devices	Disruption of Network	Thorough validation of the Input requests
WSN Node Jamming		Channel Capacity of the Network	Disruption of Data	FHSS
RF interference of RFIDs		Communication Channel	Noise in Communication Channel	Verification of the users
Sinkhole Attack	Network	Nodes of Networks	Collects the data from wireless sensor networks	Manipulate routing protocols
Man in the Middle Attack		Communication link between Node	Breach of the Data privacy	Secure Connection, VPN, End to End Encryption
IoT ransomware		IoT Cloud Platforms, IoT devices	Control devices in exchange for payment	Built in Malware Protection
IP Spoofing		IoT devices	Change the address of the device as to get data from the aunthentic source	Authentication of the IP addresses, Firewall
Phishing Attack	Application	Application layer devices	Data Leak	Avoid opening links from Unknown sources
DDoS		Security of Application Layer Device	System Disruption	Network monitoring tools
Trojan Horse		System files containing Users' data	System Hijacking	Anti-virus, Anti-malware softwares
HTTP Flooding		Web Server of the targeted system	Unable to respond to original request	Captcha Test, Web Application Firewall (WAF)
Brute Force Password Attack	Common	Secuirty software of the target system	try to breach the security measures with specific algorithm	Limit Login option, WAF
Firmware Hijacking		Parts of the Operating System	Modifies the firmware of Network	Awareness of your System
IoT Botnet		Devices of the IoT network	Disrupt the services	Update your security softwares regularly
Encryption Attack		Database of the Network	Encrypt the data at ransom	Use of strong and unique keys for Encyption

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TABLE 5 Abbreviation table.

Abbreviations	Definitions
IoT	Internet of things
SDN	Software defined network
USB	Universal serial bus
RFID	Radio frequency identification
WiFi	Wireless fidelity
E-SIM	Embedded subscriber identification module
DoS	Denial of service
LoRaWAN	Low power wide area networking
RPL	Routing protocol for low power
WSN	Wireless sensor network
FHSS	Frequency hopping spread spectrum
WAF	Web application firewall
DDoS	Distributed denial of service
HTTP	Hyper text transfer protocol
RTL	Register transfer level
IP	Internet protocol
SSH	Secure shell

Network, which includes multiple layers such as input, hiding, and output. This model has achieved an accuracy of around 98% by increasing the number of hidden layers.

2. **DDoS**: DDoS is an application layer attack in which the attacker gets complete gain over a website through dummy users. In this attack, the hacker may hijack the IoT device, and this hijacked device can make any website offline by occupying almost all of the bandwidth. Due to this attack, the attacker can access databases, application layers, and private data.[55] Asonye et al.[56] proposed the security solution against the HULK DDoS attack. The attack, which is very difficult to identify and harmful, was detected with the help of the ZigBee network.
3. **Trojan horse**: This is one of the most common attacks in which if the user downloads some malicious file, then this virus automatically gets installed inside the user PC. This virus starts collecting the information and sends the same to the attacker device. Hence, the possibility of this attack has increased due to the increase in the rate of surfing on the internet. Dong et al.[57] proposed a hardware trojan protection framework known as RG Secure, which protects by mixing the third-party IP address trusted design ideology with the scan chain netlist feature. Through this, RTL and gate-level security are achieved at around 99% accuracy.
4. **HTTP flooding**: In this attack, the attacker sends an HTTP request, which looks like the information from legal sources. There are various kinds of HTTP Flooding. Some of these attacks are carried out to target some specific sector. At the same time, there is massive HTTP Flooding, which targets many people.[58] Sangodoyin et al.[59] proposed a Software Defined Network on tree topology using Mininet emulator so that it can be used in wide area networks. This machine learning-based method gave good accuracy in detecting HTTP Flooding attacks.

3.3 | Network layer

This is the central layer in communication levels. This layer processes the data which is being collected from the sensor. This information is processed by the hub, which has well-defined algorithms.[13] There are some very common which are associated with this layer as follows:

1. **Sinkhole attack**: In this attack, the attacker captivates the information of nearby nodes and receives the packets. Now, this node transfers some data and gains access to some of the information. So, in short, it takes over all the signals

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from the wireless network and transfers some information. This attack results in the receiver not getting the message if that particular packet is stuck. Liu et al.[60] proposed a probe route-based sinkhole attack defense that protects against sinkhole attacks through far sink reverse, equal hop, and minimum hop routing.

2. **MITM attack**: This is the attack in which the hacker acts as the sender. So, the attacker, being the sender, modifies the original data and sends some different data to the receiver. Here, the attacker need not be physically present, but he can also get access to the data. For example, the hacker may get some sensors' information and send the wrong message to the system, which might switch off the system.[50] K et al.[61] proposed a logistic regression model that detects this attack through the checksum detection method and reduces it through the alternate routing method. The accuracy after this model is implemented is far better than earlier.
3. **IoT ransomware**: This is an attack in which the attacker hacks the cloud and steals the information. Now, the attacker threatens the data owner to pay money to regain the data. Sometimes, if the amount is insufficient, they may delete the data. Chakkaravarthy et al.[62] proposed an Intrusion detection Honeypot using the CEP technique for restricting some ransomware activity and gave high efficiency.
4. **IP spoofing**: This is a special attack in which the hacker enters any device through false identity. The hacker then changes the address of the packets so that the receiver device interprets it as a packet from a trusted source. After that, the packet enters the receiver device, and in this way, the hacker can steal the data very easily. Rajashree et al.[63] proposed a model that uses methods to assign IP to different devices. In this method, there are a lot of IP addresses already available. One such method, IPv4, helps exchange information between smart devices and gateway for IP assignment. Another method is IPv6, which is used as a destination option extension header.

3.4 | Common attack

Some attacks do not apply to any specific layer but may occur because of a user lack of awareness or any other reason. This type of attack is as follows:

1. **Brute force password attack**: Here, the hacker sets the algorithms which submit different passwords based on try and error. After several trials, the lock may get unlocked, and he can access data. Raikar et al.[64] proposed an approach SSH Brute force attack generation which tests the common passwords, and if it matches, then it returns a valid username and password pair. They also proposed SSH Brute force attack detection. This method helps in saving CPU, memory, and power.
2. **Firmware hijacking**: If the IoT device is not updated, then there is a high chance that the attacker may hijack the device. Sometimes, attackers may enter the device when the device is updated through an unrecognized server.
3. **IoT Botnet**: The word botnet is a combination of two words: "Bot" means robot, and "net" means network. So, in this attack, the hacker generates multiple bots simultaneously by accessing the user computer through unprotected ports. Now, this huge quantity of bots can terminate any internet-connected IoT device. These bots are commonly known as "Zombies" of the network.[65] Hussain et al.[66] proposed a two-fold machine learning approach. In the first fold, they trained state-of-the-art deep learning for detecting the IoT Botnet, while in another fold, they detected DDoS attacks. This approach gave good accuracy.
4. **Encryption attack**: Sometimes, the IoT devices are not encrypted, and the hacker may sniff the data. Once the hacker unlocks the encrypted keys, he can install his algorithm and run the device of his choice.

Table 6 shows the comparative analysis of various security attacks on IoT networks.

4 | IOT SECURITY SOLUTIONS

The security towards any attack is basic and the necessary condition.[67-70] There are various security approach which have been implemented for protection over attacks. Here are some of the security solutions which are as follows:

1. **Routing security**: Today, one of the most common and secure countermeasures is routing security. As the name suggests, we have to search the information securely. Generally, to avoid attacks, visiting a trusted website and providing

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TABLE 6 Comparative analysis of various security attacks on IoT networks.

Attack	Author	Year	Objectives	Pros	Cons
WSN Node Jamming	Ali et al.[52]	2018	Proposed different schemes like multi hop, hierarchical, trust based	Provides the protection against the WSN node jamming, black hole attack	This technique cannot be used widely becomes of some error in some field
Phishing Attack	Gopal et al.[54]	2022	Proposed an approach of Deep Neural Network model combined with autoencoder to block and prevent phishing attack	This model provide the accuracy of nearly 98.18% for multiclassification Dataset	This model is less efficient for real time datasets and in performance in various fields
DDoS	Asonye et al.[56]	2020	Provided an advanced secure network framework using different machine learning algorithm	This framework showed high efficiency in distinguishing between normal traffic and attacked traffic	This is less robust approach in real life. Moreover it may not be incorporated in cloud deployment
Trojan Horse	Dong et al.[57]	2019	Proposed a multi level trojan protection framework for perception layer in IoT known as RG Secure	Achieved up to 100% efficiency in TPR and up to 94% efficiency in TNR	This framework require more advanced SoC security for settling layout level hardware Trojan Horse
HTTP Flooding	Sangodoyin et al.[59]	2021	Proposed machine learning based detection and classification of DDoS attack	This methods of detection and classification performed well by giving good accuracy	This is not applicable for methods having control parameter as their hyperparameter have not tuned
Sinkhole Attack	Liu et al.[60]	2020	An effective probe route based sinkhole attack scheme was proposed	This approach gave better performance on defense against sinkhole attack then existing approach	In this approach the attacker is not identified and exposed at the early stages.
MITM	K et al.[61]	2022	Developed Logistic Regression Model which detects and remove the MITM	This proposed model provided an accuracy of nearly 85% against the problem of this attack	This approach is bound to handle MITM attack and several other attacks are uncovered
IoT Ran-somware	Chakkaravarthy et al.[62]	2020	Proposed an Intrusion detection Honeypot using CEP technique for restricting some ransomware activity	This approach provides high efficiency with minimum data loss against this attack	Lack of transfer learning abilities for load optimization and auto tuning feature
IP Spoofing	Rajashree et al.[63]	2018	Proposed a method for dynamic IP address assignment for preventing the IP Spoofing attack	This method uses simple algorithm and avoids using complex cryptographic schemes	In this method there are chances that the MAC address for device identifier can be duplicated
Brute Force Password Attack	Raikar et al.[64]	2021	Proposed an IDS based attack detection and mitigation method	This method saves around 25% of CPU, 10% of memory and 40% of power	This method does not handle the insider attacks in IoT networks
IoT Botnet	Hussain et al.[66]	2021	Proposed a two fold machine learning approach for prevention and detection of IoT Botnets	This model gave persistently good accuracy and robust to prevent and detect attack	This model is not implemented on live network traffic

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personal data to websites with secure protocols is safe. Using different paths is a good practice for attaining security over the sensor network.

2. **End to end encryption**: For the communication to be secure, it is essential to note that it must be encrypted. The encrypted mode of communication is the safe path, as the decryption key is present only at the receiver. The benefit of this technique is that the person without the key cannot decrypt the message; hence, data privacy will be maintained.
3. **Data integrity**: Whenever the hacker injects malicious code inside any application, it creates errors. To prevent this, the receiver device should be able to detect the error inside. Various error detection methods include parity checker, Hamming distance, Checksum, CRC, and so forth. So, with the help of this technique, we can easily detect the error. So whenever this error is detected, the system automatically drops that information packet.
4. **Tamper proofing**: The tamper-proofing technique is generally used to prevent the access of any third party over the device or the data. Through this technique, the issue of IoT device tamper problem is solved. This is implemented with the help of tamper resistance elements installed on the IoT device chip.[71]
5. **Fragmentation redundancy scattering**: The Fragmentation Redundancy Scattering generally stores sensitive and vital data on the cloud. Here, the data is fragmented into various parts. Now, these fragmented parts are stored on different servers. So if servers get hacked, only that part of the data will be leaked, and the rest of the information will still be secure.
6. **Anti virus**: This is important in preventing the virus from entering the user device. The antivirus application has a list of codes for various viruses. Whenever it runs a scan and a virus matches the antivirus code, it immediately throws that file or alerts the user. The most important thing is that the user should always keep the antivirus in updated form because whenever any new virus comes into the market, the company will immediately add it to the application. Hence, it can detect new viruses also.

5 | FUTURE CHALLENGES AND RESEARCH OPPORTUNITIES

Today, security is the most essential part of communication. Any information that is traveling from sender to receiver must be in encrypted form so that a third party will not access the information.[72-74] Moreover, data security in IoT is not easy because the IoT network is very vast, so it is a tedious task to distinguish between original and duplicate information. So here we have discussed some security issues related to IoT. They are as follows:

1. **Data encryption**: As we discussed, the IoT network is vast; the IoT devices have to store and transfer a large amount of data. So, if anyone can attain access to the transmission path during the transmission, he can steal the information before it reaches the receiver. So, to prevent this, the data needed to be sent in an encrypted mode so that anyone accessing the information would be unreadable. In encryption, if it is not applied properly, there is a high chance that the attacker can read the information. So the information needed to be properly encrypted.[75]
2. **Hijacking of IoT devices**: Today, as the number of IoT devices has increased, the attacks belonging to these devices have also increased. One of the major issues with IoT devices is that in the absence of antivirus or any malware protection software, these devices can be easily hijacked, leading to a major issue. For example, a self-driving car contains an IoT device, and if it is hijacked, then there are chances that it loses its basic functionality and creates a big loss. However, this issue is slowly getting solved as the data is stored in the cloud, but it is not solved completely.[76]
3. **Financial crime through IoT**: Nowadays, there are lots of ways to get control over IoT devices. For example, an attacker may get control over any camera that records your daily routine. Then he asks for a handful of money. Hackers also use many other ways to perform this kind of activity. However, these scams are slowly decreasing due to increased awareness and an increasing understanding of deep learning concepts among educated people.
4. **Access regulation and authorization**: Authorization dictates that a device or user can access specific resources after being identified once. Through Access Regulation, resources are granted or denied following the criteria. The criteria for access to the resources have to be easier to create, simple, and easy to manipulate to achieve Data Confidentiality.[77]
5. **Identity administration and authentication**: With the need for technologies and algorithms that can administrate profiles and authenticate the transaction of data of the user, there are many research issues in IoT as multiple devices and users connect and share data after authenticating each other using trusted services. As all these things have to be identified uniquely, these administrative methods must be researched profoundly.[78]
6. **End-to-end security**: Data security in communication at endpoints of IoT devices and Internet hosts is important. To completely secure both ends, verifying both individuals, proper algorithms, and protocols must be implemented

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and improved periodically for any new threats. In communication with end-to-end security, both sender and receiver can rest assured that their communication is not public to anyone else, and no one other than them can interfere with their information. So, nowadays, proper end-to-end security is a basic necessity, and if not, then many applications would not be able to work smoothly.[79]

6 | CONCLUSION

In this paper, we conducted an extensive survey on IoT technologies such as ZigBee, RFID, Bluetooth, WiFi, and so forth, IoT-layered architecture, classified into perception, network and application layers. Next, we have highlighted various communication protocols considering the layers of the IoT-layered architecture. Furthermore, we have discussed several security attacks based on the IoT-layered architecture. Considering the security attacks, we have presented IoT security solutions along with the possible open research issues and opportunities that can impact the security of IoT networks. In the future, we will incorporate a decentralized and secure platform to improve the security of IoT networks in the real-time and dynamic scenarios. Further, we will implement machine learning and deep learning techniques with blockchain technology to secure the IoT networks accurately. Blockchain and AI techniques help to improve the IoT networks communication further securing the data efficiently and reliably.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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How to cite this article: Patel A, Patel D, Kakkar R, et al. Safeguarding the IoT: Taxonomy, security solutions, and future research opportunities. *Security and Privacy*. 2024;7(2):e354. doi: 10.1002/spy2.354

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